

BASIC SIGHT-SINGING AND EAR TRAINING FOR CHURCH SINGERS

UNIT 6.2 – PITCH THE MINOR SCALES – NATURAL AND HARMONIC. ACCIDENTALS.

THE KEYS OF A MINOR,
E MINOR, AND D MINOR.



MODES AND MINOR SCALES

When we first introduced the concept of the **tonic**, (Unit 3.2, Slide 2) we mentioned that in the “movable DO” system the **tonic note**, or the **point of reference** is typically called “DO.” This is true in the case of melodies built on a **major scale**, that is to say, in a **major key**.

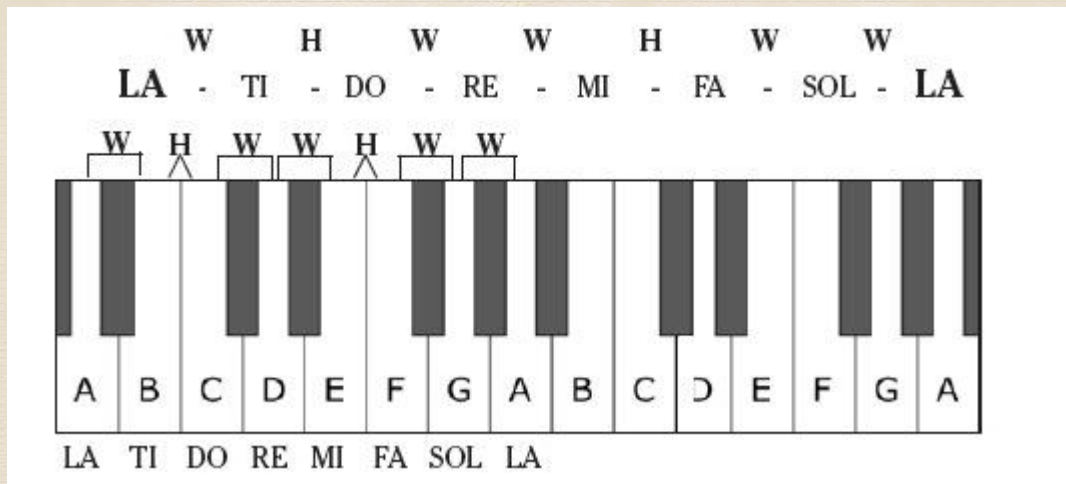
But we can also build scales on other starting notes besides DO, generating what are known as **modes**, such as Dorian (starting on RE), Phrygian (starting on MI), Lydian (starting on FA), and Mixolydian (starting on SOL). These scales or modes have different combinations of whole and half steps. Each combination of whole-steps and half-steps had its own distinct sound and **emotional “feel”** or “**flavor**,” although the exact nature and character of that emotion is somewhat difficult to characterize.*)

One of the most frequently encountered modes in liturgical chant is the **Aeolian** mode, with a tonic on LA, otherwise known as the **natural minor scale**.

*) The Ancient Greeks believed that music in various modes had a specific effect on the emotions and even the morals of individual listeners and even entire societies (see the writings of Plato and Aristotle).

THE NATURAL MINOR SCALE

The natural minor scale, by definition, has the following sequence of intervals:

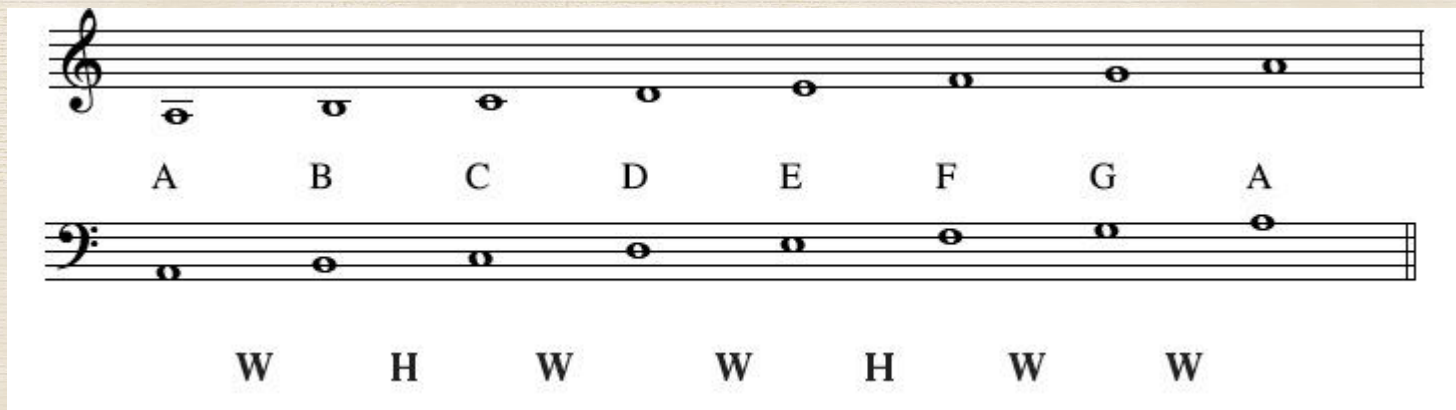


It is particularly important to note the location of the half steps—after the second note, B = TI, and after the fifth note, E = MI. These are the notes that impart to the A minor scale its **minor character**—its “minor-ness.” Compare this with the major scale, which in both cases has a whole step following the second and fifth note.

THE NATURAL A MINOR SCALE

To write the natural minor scale in staff notation, we begin on the note “A” and use only the notes corresponding to the white keys of the piano (just as the C-major scale starts on the note “C” and uses only the white keys). This scale is called the natural A Minor, and because it uses the same notes as C Major, it is referred to as the **relative minor** of C Major. The **key signature** of both A Minor and C Major are identical: no sharps or flats.

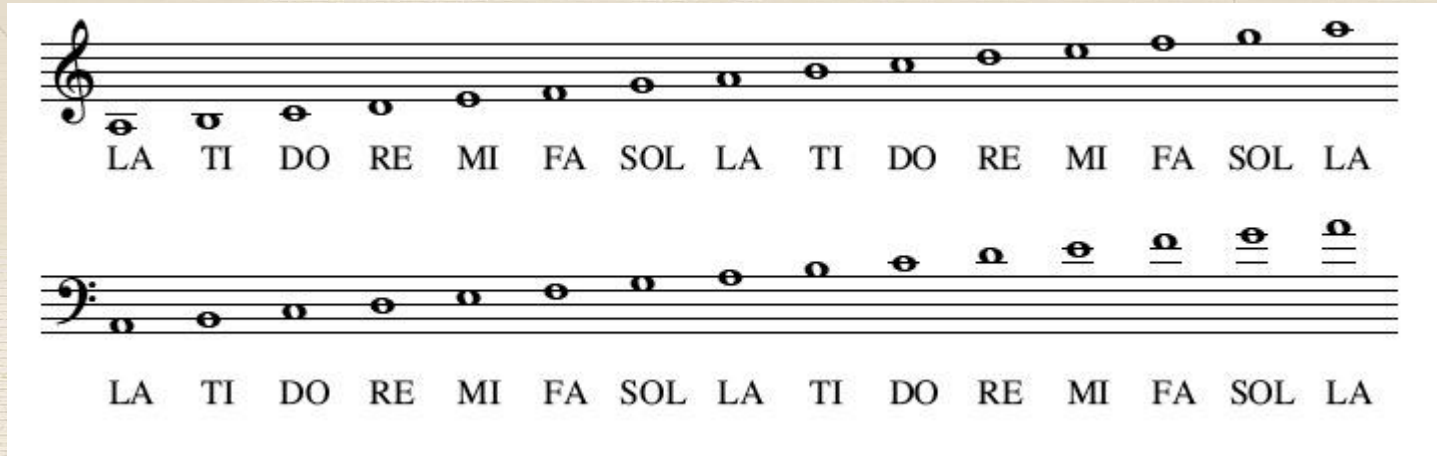
Here is the natural A Minor scale in treble and bass clef. Note the arrangement of whole-steps and half-steps:



The image displays the natural A minor scale on two staves. The top staff uses a treble clef and the bottom staff uses a bass clef. Both staves show the notes A, B, C, D, E, F, G, and A. Below the notes, the intervals between them are labeled: W (Whole) between A and B, H (Half) between B and C, W (Whole) between C and D, W (Whole) between D and E, H (Half) between E and F, W (Whole) between F and G, and W (Whole) between G and A.

Staff	Clef	Notes	Intervals
Top	Treble	A, B, C, D, E, F, G, A	W, H, W, W, H, W, W
Bottom	Bass	A, B, C, D, E, F, G, A	W, H, W, W, H, W, W

Here the natural A-minor scale is shown in a two-octave range, used by singers. Use both the SOL-FA syllables and the notes to practice the intervals:



Please refer to the Pitch Supplement audio files (6.2-1, 6.2-2, 6.2-3, and 6.2-4) in the Week 6 Materials section of Google Classroom. As you sing along, point to the notes on this slide.

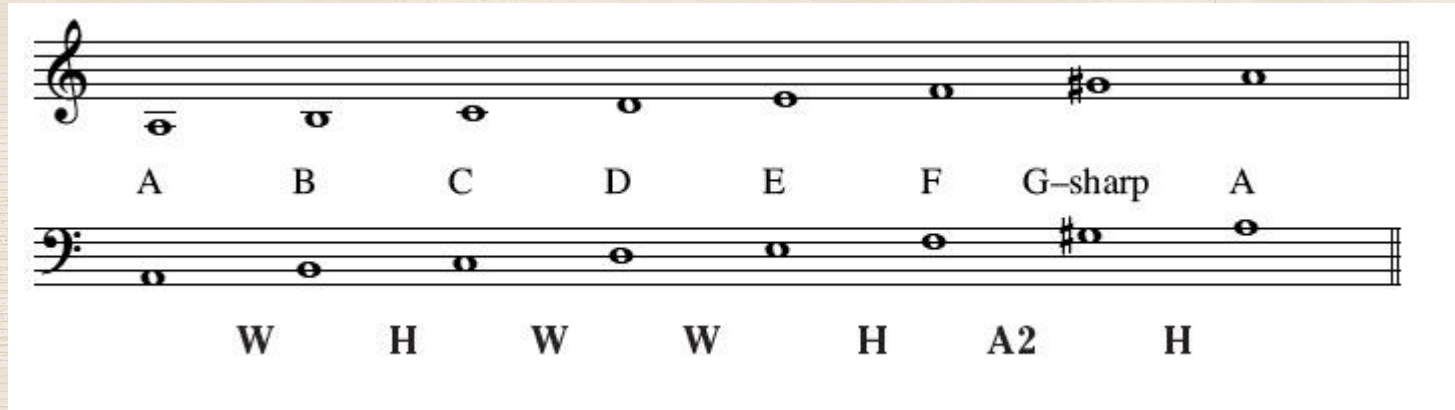
THE HARMONIC A MINOR SCALE

Because of the strong melodic attraction between the tonic and the note immediately below it (called the **leading tone**, because it “leads” the ear to the tonic), the seventh note of the natural minor scale—“SOL”—is frequently found to be raised half a step by the addition of a **sharp** (#). According to the terminology of solfege, SOL-sharp is called “SI.” The resulting scale is called the **harmonic minor scale**. By definition, it has the following sequence of intervals:

	W		H		W		W		H		A2		H	
LA	-	TI	-	DO	-	RE	-	MI	-	FA	-	SI	-	LA

Notice a previously unseen interval—A2—the “**augmented second**”—so called because the interval from FA to SI is a half step **larger** than the major second FA to SOL. It has, in effect, been “augmented” by the addition of the sharp.

In staff notation, the harmonic A Minor scale can be written, like the natural minor, starting on the note “A.” but once the note “G” is reached, a sharp must be added. Since this sharp is not in the key signature of A, it is written next to the note itself . For this reason we refer to it as an **accidental** (that is, it affects only the note next to which it is written and, in music with measures, other identical notes in that measure).



Music written in the key of A Minor will frequently contain G-sharps. This is one way of telling whether a piece with no sharps or flats in the key signature is in C Major or in A Minor: if it's in C Major, it will likely not have any accidental G-sharps.

ACCIDENTALS IN THE MOVEABLE-DO SYSTEM

The Movable-DO system accommodates accidentals by altering the name of the SOL-FA syllable to which an accidental is applied. Sharps are expressed by changing the vowel of the SOL-FA syllable from “-A,” “-E,” or “-O” to “-I.”

DO-sharp becomes **DI**

RE-sharp becomes **RI**

FA-sharp becomes **FI**

SOL-sharp becomes **SI**

LA-sharp becomes **LI**

Notice that the relationship of a “sharped” syllable (DI, RI, FI, SI, LI) is the same to the note that follows it as the familiar **MI** to **FA** and **TI** to **DO**, that is, a **half step**.

The names of notes with accidental flats in the Movable-DO terminology are formed in a similar way, by altering the vowel of the SOL-FA syllable: “-E” changes to “-A,” while “-A,” “-I” and “-O” change to “-E.” Thus

RE-flat is called **RA**

MI-flat is called **ME**

SOL-flat is called **SE**

LA-flat is called **LE**

TI-flat is called **TE**

Flat accidentals are not as common in harmonized Orthodox church music. They are encountered quite extensively in certain modes of Byzantine, Serbian, and Carpatho-Rusyn chant.

OTHER COMMON MINOR SCALES – D MINOR AND E MINOR

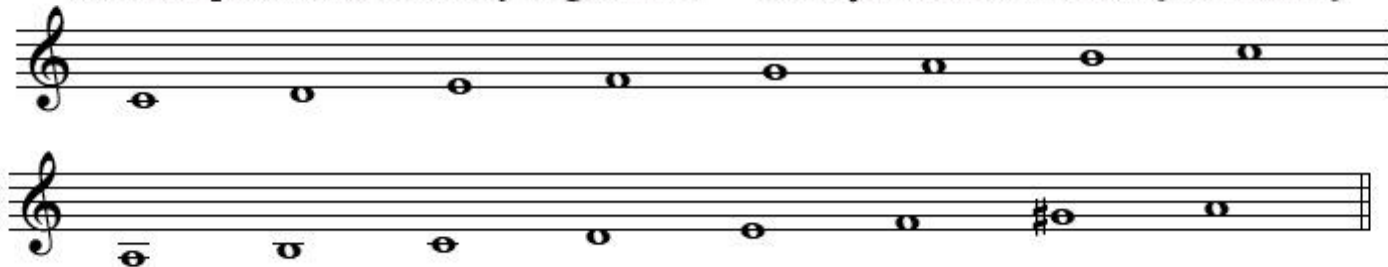
As with the major scale, various other minor scales (both natural and harmonic) can be formed, using SOL-FA syllables, using **any** pitch as the starting pitch and calling it “LA.”

Just as we formed the **relative minor** of C Major—A Minor—by going down DO-TI-LA, from the note C to the note A, we can find the relative minor keys of other common keys, such as F Major or G Major, by singing down DO-TI-LA—from the note F to the note D, or from the note G to the note E. We can then build minor scales starting on D or starting on E.

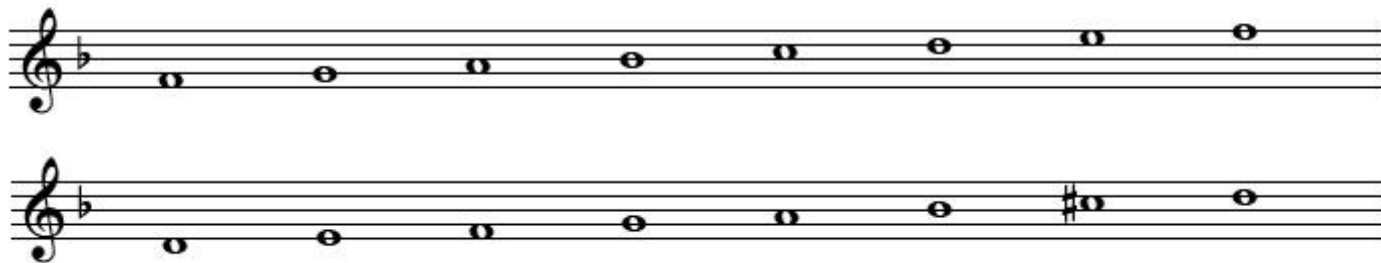
Just as the tonic of a major scale is always called DO, the **tonic of a minor scale is always called LA**, and relationships between the notes (i.e., the intervals, such as LA-DO, TI-RE, LA-MI, etc.) remain consistent. The only thing that changes is the tonic or reference pitch.

THREE COMMON MAJOR SCALES AND THEIR RELATIVE MINOR COUNTERPARTS

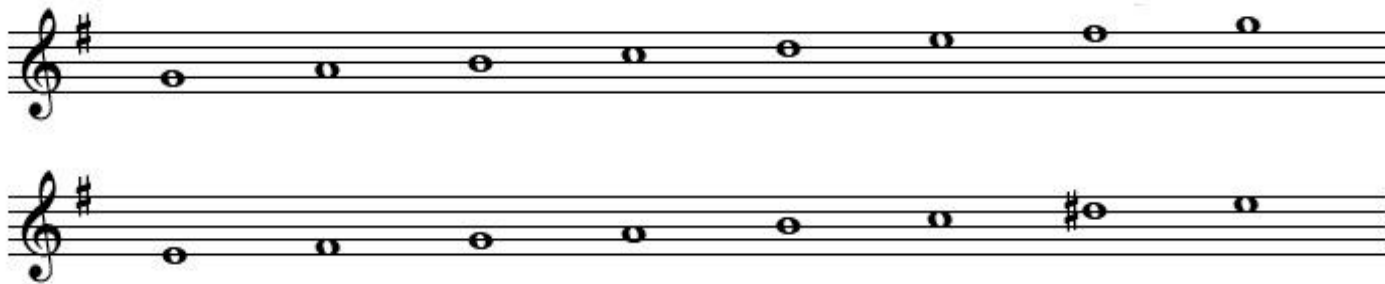
No Sharps or Flats in Key Signature — C Major and A Minor (harmonic)



One Flat in Key Signature — F Major and D Minor (harmonic)



One Sharp in Key Signature — G Major and E Minor (harmonic)



“ON THE SPOT” EXERCISE

1. Using your knowledge of SOL-FA syllables, sing each of the above scales.
2. In each scale, identify and write down the notes that form intervals of:
 - a. Major thirds
 - b. Minor thirds
 - c. Perfect fourths
 - d. Perfect fifths
3. Look for parallels and correlations: What SOL-FA syllables are used to name these intervals in the various scales?

Please proceed to
**PITCH EXERCISES AND
SUPPLEMENTS 6.2**

